Evaluating a Random Forest-based System for Interpreting Attrakdiff Quantitative UX Questionnaires

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ABSTRACT

This paper explores the design and evaluation of a machine learning artefact aimed at supporting the interpretation of quantitative user experience questionnaires results. While these methods are widely used due to their scalability and reliability, they primarily produce numerical scores that require additional effort to translate into actionable design insights. This interpretative step remains time-consuming and dependent on expert judgment. To address this limitation, we propose an artefact that leverages a supervised machine learning model to automatically map quantitative questionnaire results to concrete interface improvement categories. Specifically, we investigate the use of a Random Forest classifier trained on paired data consisting of AttrakDiff numerical scores and the set of feedback directly provided by the testers. We also design an experimental study to compare the results of the machine learning artefact with human-generated feedback, in order to assess whether the model correctly predicts relevant categories of improvement. In this evaluation process, participants interact with a web interface, complete an AttrakDiff questionnaire, and then provide explicit recommendations on what should be improved. These responses are used as a reference to evaluate the model's predictions. Additionally, we rely on standard machine learning metrics to assess the model's performance, such as precision, recall, and F1-score, which provide a more detailed understanding of how accurately the system predicts the appropriate categories.

Author Keywords

User Experience; UX Evaluation; Quantitative Questionnaires; Machine Learning; Artificial Intelligence; Interface Design

INTRODUCTION

User experience evaluation methods play a central role in the design and development of interactive systems. They enable designers to assess how users perceive, understand, and engage with an interface. Among these methods, standardized questionnaires such as the User Experience Questionnaire (UEQ) or the System Usability Scale (SUS) are widely adopted due to their scalability, ease of deployment, and ability to produce reliable quantitative measurements across large user groups. Despite their widespread use, quantitative UX questionnaires present an important limitation. While they provide structured numerical insights into users perceptions, they do not directly indicate which specific elements of an interface should be modified. The process of translating these numerical results into actual areas of modification within the interface is often manual, time-consuming, and highly dependent on the expertise of UX practitioners, making it difficult to scale or standardize. As highlighted in prior work, the gap between quantitative UX evaluation and concrete design recommendations remains relatively underexplored. While existing research has focused extensively on validating questionnaire instruments or analyzing user behavior, fewer contributions address how numerical UX data can be interpreted and transformed into actionable insights. At the same time, machine learning techniques are increasingly present in HCI for tasks such as user modeling, personalization, and interaction prediction. However, their application to the interpretation of UX questionnaire results remains limited, particularly in the context of interpretable models capable of handling structured data. To address this gap, this paper proposes a novel artefact : a machine learning–based system designed to translate quantitative AttrakDiff questionnaire results into structured categories of interface improvements. Specifically, the system relies on a supervised learning approach based on a Random Forest classifier, chosen for its robustness on structured data, its ability to capture non-linear relationships, and its relative interpretability compared to more complex models. This model maps quantitative questionnaire scores to actionable categories of interface improvements. These categories are defined beforehand based on a dataset of previously collected AttrakDiff responses, combining numerical scores with user-provided feedback on concrete interface issues. While a detailed description of these categories is provided later in the paper, they broadly cover aspects such as aesthetics, creativity, and professionalism. The goal at this stage is to define categories that are sufficiently general to capture a wide range of important UX dimensions, while still being informative enough to guide the interpretation of questionnaire results.

Accordingly, this work addresses the following research question :

How accurately can a Random Forest model translate quantitative AttrakDiff questionnaire results into action-

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able categories of interface improvements, compared to human-generated design feedback ?

RELATED WORKS

Machine Learning for UX Evaluation

The existing literature related to this research topic explores a wide range of applications of machine learning and artificial intelligence in the field of user experience. These applications vary significantly depending on the stage of the UX process in which they are introduced, including evaluation, modeling, and adaptation of user interfaces.

One line of work focuses on augmenting UX evaluation through implicit user signals. For example, Koonsanit et al. [2] propose a framework to classify user satisfaction based on facial expressions, combined with demographic features such as age and gender. The main contribution of this work lies in reducing reliance on self-reported questionnaires by leveraging automatically captured behavioral data. Their approach consists of collecting facial expression data during interaction, transforming these signals into numerical features, and training several supervised learning models including Support Vector Machines, K-Nearest Neighbors, Logistic Regression, and Neural Networks.

The evaluation of their framework is conducted through multiple experimental steps. First, preliminary experiments validate the facial expression recognition system. Then, a dataset is constructed by combining time-series facial expression data with self-reported satisfaction scores collected through questionnaires. The models are evaluated using standard classification metrics such as accuracy, precision, and recall, and further validated using cross-validation techniques (e.g, leave-one-out cross-validation). Their results show that user satisfaction can be predicted with relatively high accuracy (up to 86%). However, the output remains limited to overall satisfaction prediction and does not provide detailed insights into specific interface issues.

Interpreting Quantitative UX Questionnaires

Another line of research that is closer to our studied subject investigates how to better interpret structured UX evaluation questionnaires. A notable example is the work of Hinderks et al. [1], which combines the User Experience Questionnaire (UEQ) with Importance-Performance Analysis (IPA). The main contribution of this work is to enhance the interpretability of questionnaire results by transforming raw UX scores into structured, visually interpretable insights. More specifically, UX dimensions measured by the UEQ (e.g attractiveness, efficiency, stimulation) are treated as attributes and mapped onto a two-dimensional space representing perceived importance and performance. The proposed method consists of three main steps. First, UX dimensions from the UEQ are selected as attributes for the analysis. Second, both performance (directly derived from UEQ scores) and importance (collected through additional questionnaire items) are obtained for each attribute. Finally, these values are plotted in an IPA graph, where each dimension is positioned according to its importance and performance. The resulting visualization is divided into four quadrants (e.g., "Keep Up the Good Work", "Concentrate Here"), each associated with a specific type of design recommendation.

To validate their approach, the authors conducted a study with 219 participants evaluating two widely used platforms, YouTube and WhatsApp. Participants completed different versions of the UEQ, allowing the authors to compute both performance and importance scores for each UX dimension. The reliability of the collected data was assessed using measures such as Cronbach's alpha to ensure internal consistency of the questionnaire scales. The resulting IPA plots enabled the identification of UX dimensions requiring improvement, demonstrating that the method can provide more actionable insights compared to standard questionnaire analysis. However, this approach primarily focuses on structuring and prioritizing UX dimensions rather than detailing the specific nature of the underlying interface issues. It provides a valuable way to better interpret questionnaire results by highlighting which aspects of the user experience deserve attention, offering designers clearer guidance compared to raw scores alone. In contrast, the present work explores a complementary direction by focusing on the nature of the issues themselves. Instead of emphasizing the relative importance of UX dimensions, our approach aims to translate quantitative AttrakDiff results into more concrete categories of interface improvements (e.g aesthetics, creativity), this way moving towards a more design-oriented interpretation of questionnaire data.

Machine Learning for UX Dimension Classification from User Feedback

More recent work has explored the use of machine learning to structure qualitative user feedback into UX-related dimensions. For instance, a recent study conducted by Ramjul et. al [3] proposes a system that classifies mobile application user reviews into UX dimensions derived from the UEQ-S framework. The contribution of this work lies in transforming unstructured textual feedback into structured UX categories such as clarity, efficiency, and originality. The evaluation strategy consists of training and comparing multiple supervised learning models, including Logistic Regression, Support Vector Machines, Random Forest, and BERT, using standard metrics such as precision, recall, F1-score, and accuracy. Results show that while BERT achieves the highest performance (F1 = 76%), Random Forest remains competitive, offering a strong trade-off between performance and interpretabiliity. This work is particularly relevant to our idea, as it demonstrates the feasibility of mapping user feedback to structured UX dimensions using machine learning. However, it differs in that it relies on textual input data, whereas our work focuses on structured quantitative questionnaire results (AttrakDiff). Furthermore, while their system classifies existing feedback, our proposed artefact aims to infer actionable interface improvement categories directly from numerical evaluation data.

CONSTRUCTIVE CONTRIBUTION

This work proposes an artefact based on a Random Forest model to support the interpretation of quantitative UX questionnaire results. The core contribution consists of a system

that maps AttrakDiff scores to structured categories of interface improvements, transforming abstract numerical evaluations into more actionable design insights. Rather than attempting to directly predict highly specific interface modifications, the system operates at an intermediate level of abstraction by identifying categories of improvements such as clarity, intuitiveness, aesthetics, and creativity. This design choice ensures that the output remains both generalizable across different interfaces and useful for guiding design decisions. This study therefore has two main contributions :

- the construction of a dataset linking quantitative UX evaluations to qualitative design recommendations
- the development of a supervised learning model capable of learning these relationships and generating structured predictions

First Prototype and Research Focus

The first prototype focuses on the feasibility of learning meaningful mappings between quantitative AttrakDiff scores and categories of interface improvements. It is implemented as a supervised multi-label classification system based on a Random Forest model trained on structured UX evaluation data. This prototype addresses the following sub-question : **To what extent can a Random Forest model learn consistent relationships between AttrakDiff scores and high-level categories of interface issues ?**

Regarding the role, implementation, and look and feel triangle, this prototype primarily focuses on the implementation dimension, as it investigates the technical feasibility of mapping questionnaire results to categories of interface issues. It also addresses the role dimension by exploring how such a system could support UX practitioners in interpreting evaluation results. The look and feel dimension is less central at this stage, but it is still considered in terms of how practitioners would interact with the system in practice, for example through a simple web-based interface allowing them to input AttrakDiff data and visualize the model's output in a clear and efficient way.

Data Preparation

The first step in building the artefact consists of constructing a dataset that links quantitative AttrakDiff evaluations to concrete interface improvement categories. To achieve this, we rely on previously conducted UX evaluations of various interfaces and systems, collected through a literature review of research papers and case studies where AttrakDiff was used alongside qualitative analyses or expert recommendations. To ensure that the model can learn meaningful and generalizable patterns, the dataset must include a sufficient number of evaluation instances. In this work, we aim to collect at least around one hundred questionnaire responses paired with qualitative feedback, providing a minimal but workable basis for training the model. Importantly, these evaluations do not need to concern the same interface, as the objective is to capture generalizable relationships across different contexts.

For each selected study, two types of data are extracted :

- **Quantitative data** : vectors of numerical AttrakDiff scores
- **Qualitative data** : reported conclusions or feedback describing which aspects of the interface should be improved

The qualitative feedback is manually analyzed to define a set of recurring categories of interface improvements, which serve as labels for the model. These categories are selected as they represent broad and commonly used dimensions for evaluating interactive systems, allowing the model to capture general UX issues across different contexts. The selected set includes :

- clarity
- intuitiveness (ease of navigation and interaction)
- aesthetics (visual appeal, colors, layout)
- creativity/originality (novelty and engagement)
- professionalism (consistency and perceived quality)
- completeness (coverage of features or information)

These categories are then encoded into a structured format where each instance is represented by numerical questionnaire scores associated with one or multiple binary labels, resulting in a multi-label classification problem.

Training of the Model

The artefact relies on a supervised learning model trained to map questionnaire scores to interface improvement categories. A Random Forest classifier is used, as it is particularly well-suited for structured data and offers a good balance between performance, robustness, and interpretability. A Random Forest is an ensemble method that builds multiple decision trees during training. Each tree is trained on a randomly sampled subset of the dataset while also considering a random subset of features at each split. This introduces variability among the trees, and the final prediction is obtained by aggregating their outputs, typically through majority voting. This process helps reduce overfitting and enables the model to capture complex, non-linear relationships between questionnaire dimensions and design-related categories.

Before training, the dataset is split into training and testing subsets to ensure that the model generalizes to unseen data. The training set is used to learn the mapping between input score vectors and output categories, while the test set is reserved for evaluating performance. Once trained, the model takes as input a vector of AttrakDiff scores and outputs one or several predicted categories of interface improvements, providing structured guidance for UX analysis.

Use of the Artefact in Practice

Once the model is trained, it can be used as a support tool for UX practitioners during the analysis phase of user evaluations. In a typical scenario, a UX designer conducts a user study on a given interface and collects quantitative data using the AttrakDiff questionnaire. These questionnaire results are then provided as input to the system in the form of a vector of scores. Based on this input, the trained model generates a set of predicted categories of interface improvements among the ones presented earlier. The UX practitioner can then use these outputs to guide further analysis by linking the predicted

categories to concrete interface elements. For example, a predicted issue related to aesthetics may lead to examining colors or layout.

The sketch in Figure 1 illustrates this overall process.

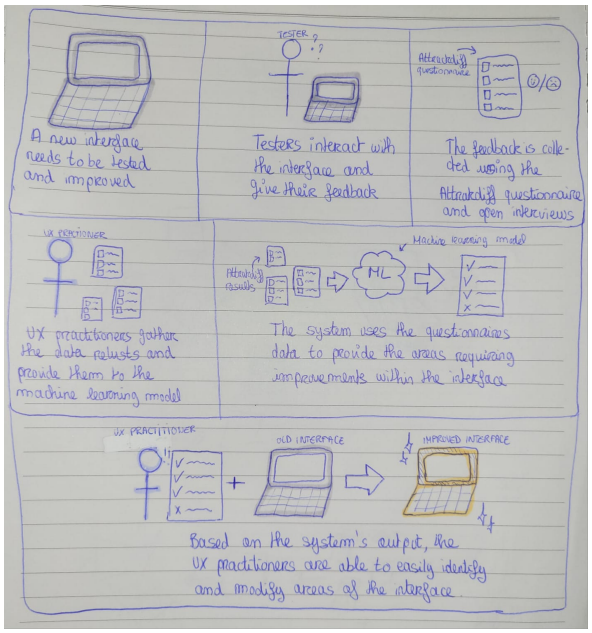


Figure 1. Overview of the system process

EVALUATION

To assess the proposed artefact, we design a two-part evaluation strategy combining a technical evaluation of the model's predictive performance and a comparative study against human-generated design feedback.

Technical Evaluation of the Model

The first evaluation focuses on the predictive performance of the Random Forest model as a multi-label classifier. The dataset constructed during the data preparation phase is divided into training and testing sets, specifically a 80/20 split. The model is trained on the training set and evaluated on unseen test data. Performance is measured using standard multi-label classification metrics :

- **Precision** : proportion of correctly predicted categories among predicted ones
- **Recall** : proportion of relevant categories correctly identified
- **F1-score** : harmonic mean of precision and recall
- **Accuracy** : proportion of exact matches between predicted and true category sets

These metrics provide insight into how well the model captures the relationship between questionnaire scores and interface improvement categories.

Human Comparison Study

To evaluate how well the model's predictions align with human judgments, we present a controlled user study with participants from diverse backgrounds to ensure a variety of perspectives. We recruit approximately a dozen participants, including UX designers, software engineers, design students, computer science students, and general end-users. This diversity allows us to examine how different types of users perceive the same interface and whether the model can reflect these differences in its predictions.

All participants interact with the same web application : a publicly available task management app similar to Trello, featuring boards, lists, and cards. They are asked to complete a set of predefined tasks to ensure consistent interaction across participants, for example :

- Create a new project board and add three tasks with due dates
- Assign tasks to a user and move them across lists to reflect progress
- Edit task details, including labels and descriptions, and delete a task
- Use the search and filtering functionality to find specific tasks

After completing these tasks, participants fill out an AttrakDiff questionnaire evaluating their expeirience and they are shown the set of improvement categories previously defined in the study for them to choose which areas are the most problematic within the interface.

The AttrakDiff responses from each participant are fed into the trained Random Forest model to predict the categories of interface improvement. By comparing the model's predictions with the participants' qualitative categorizations, we can assess whether the system could adapt to different user perspectives and successfully capture variation in perceived interface issues depending on user background. This setup allows us to verify not only the general accuracy of the model but also its sensitivity to diverse user profiles and subjective interpretations of interface quality.

DISCUSSION

The proposed evaluation combines a technical assessment of model performance with a comparison against human-generated feedback. While this approach provides a meaningful first validation of the artefact, several limitations must be acknowledged.

First, the validity of the model is strongly dependent on the quality and size of the dataset used for training. Since the dataset is constructed from previously published UX evaluations and manually interpreted qualitative feedback, it reflects subjective judgments made by researchers or participants. The process of defining categories from this data also introduces an additional layer of interpretation, which may influence the consistency and reliability of the labels. As a result, the model does not learn an objective ground truth, but rather a structured representation of human interpretations.

Second, the human comparison study relies on the assumption that user-generated feedback can serve as a reliable benchmark. However, user feedback itself can be inconsistent, incomplete, or influenced by individual differences in expertise and perception. While agreement between the model and users can indicate meaningful alignment, disagreement does not necessarily imply incorrect predictions, but may instead reflect ambiguity in UX interpretation.

Despite these limitations, the proposed approach demonstrates potential for generalisation across different types of interfaces and evaluation contexts. By operating at the level of abstract categories, the system is not tied to a specific interface or domain. This abstraction enables the model to capture patterns that are transferable across different UX scenarios. However, this generality also introduces a trade-off. While category-level predictions are broadly applicable, they remain relatively high-level and do not directly identify specific interface elements that should be modified. In this regard, the artefact should be understood as a decision-support tool that narrows down the space of possible improvements rather than providing precise design prescriptions. The generalisability of the model is also dependent on the diversity of the training dataset. A dataset covering a wide range of interfaces, interaction types, and user populations would improve the robustness of the learned patterns. Conversely, a limited or biased dataset may restrict the applicability of the system to similar contexts.

Overall, while the current artefact remains a high-level interpretative tool, it demonstrates the feasibility of leveraging machine learning to support the translation of quantitative UX evaluations into structured insights. This represents a step toward more integrated and data-driven UX analysis workflows.

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Explain how generative AI has been used in the assignment:

For each instance of use (see examples above), write an entry below and explain
1) *Purpose of use (what did you use the tool for?)*
2) *Which tool did you use? Include the link if possible, e.g., [Copilot with enterprise data protection (UCPH license), https://copilot.microsoft.com]*
3) *Work phase or step (when in the process did you use GAI?)*
4) *What did you do with the output? (including any editing of or continued work on the output)*
5) *What did you benefit of the tool? (think about the usefulness of the tool and reflect on your experience of using it, the applicability of the tool for this use, and your skill development in using GAI tools)*

1) I used ChatGPT to reformulate some technical words and concepts from the articles I read, that were not very clear to me. I also provided ChatGPT with some phrases and parts of my work where I wasn't sure of the translation and the overall structure and grammar.
2) ChatGPT
3) Mainly at the end when I re-read my work and wanted to make sure that the text was grammatically correct and clear.
4) I usually copy the output, read it and modify it to make it more fluid with the rest of my text.
5) The tool allow me to have a better understanding of some concepts and also to improve my English writing.

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Figure 3. Declaration AI Page 2

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- *Explanation of concepts, definitions and models for better understanding*
- *Summarising large amounts of information*
- *Help with structure and outline*
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- *Data processing (e.g. transcription, cleaning and structuring data)*
- *Data analysis*
- *Grammar check*
- *Feedback / review tool for revising text*
- *Grammar / language check tool*
- *As object of the analysis*

Figure 2. Declaration AI Page 1